Chapter 3. Elementary Counting Problems

There Are a Lot of Them

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Pigeon-holes and induction draw conclusions *from* numbers; this chapter develops the basic techniques for *finding* them.

3.1 Permutations

Motivation. Order distinct objects; handle repeats by labelling them and dividing out the labellings.

Definition 3.1 (Permutation, factorial)

An arrangement of distinct objects in a linear order, using each exactly once, is a **permutation** of them. The number $n! := n(n-1)\cdots 2\cdot 1$ of permutations of n objects is n **factorial**.

Theorem 3.2

An n-element set has $n!$ permutations.

Proof. n choices for the first position, for each of them $n-1$ for the second, …, 1 for the last. ■

By convention $0! = 1$. **Stirling's formula** (without proof):

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n, \qquad \text{i.e.} \quad \lim_{n\to\infty} \frac{n!}{\sqrt{2\pi n}(n/e)^n} = 1. \tag{3.1}$$

Example 3.3

Flags with three horizontal stripes of the distinct colours red, white, green: $3! = 6$ (RWG, RGW, WRG, WGR, GWR, GRW).

Example 3.4

Five red, three yellow and two white flowers can be planted in a row in $A = \frac{10!}{5!\,3!\,2!} = 2520$ ways.
Solution. Labelling the flowers of each colour, each pattern gives $5!\,3!\,2!$ of the $10!$ labelled rows: $A \cdot 5!\,3!\,2! = 10!$.

A collection with repetitions is a **multiset**; a linear order of all its elements is a **multiset permutation**.

Theorem 3.5 (Multiset permutations)

Let $a_1 + \cdots + a_k = n$ be nonnegative integers. A multiset with a_i objects of type i ($i \in [k]$) has

$$\frac{n!}{a_1!\,a_2!\cdots a_k!}$$

linear orders.

Proof [Supplement]. Label the objects of type i by $1, \ldots, a_i$. Each permutation of the n labelled objects arises from exactly one pair (linear order, labelling), and a fixed order has $a_1!\cdots a_k!$ labellings. So $A \cdot a_1!\cdots a_k! = n!$ (Theorem 3.2). ■

3.2 Strings over a finite alphabet

Motivation. Words count functions, words without repeats count injections, and bijections transfer counts.

Theorem 3.6

There are n^k strings of length k over an n-element alphabet.

Proof. Each of the k positions can independently be filled in n ways (letters may repeat). ■

Example 3.7

There are $10^k - 10^{k-1} = 9 \cdot 10^{k-1}$ positive integers with k digits.

The bijective method. Pair the elements of two sets; if nobody is left over, the sizes agree.

Definition 3.8 (Bijection)

Let X, Y be finite sets and $f : X \to Y$ a function such that (1) $f(a) = f(b) \Rightarrow a = b$, and (2) every $y \in Y$ equals $f(x)$ for some $x \in X$. Then f is a **bijection** from X onto Y; equivalently, each $y \in Y$ has exactly one preimage.

Definition 3.9 (Injection, surjection)

A function satisfying (1) is **one-to-one** (**injective**); one satisfying (2) is **onto** (**surjective**).

Proposition 3.10

If there is a bijection from a finite set X onto a finite set Y, then $|X| = |Y|$.

Proof. The pairs $\{x, f(x)\}$ partition $X \cup Y$ with one element from each set; if there are m pairs, $|X| = |Y| = m$. ■

Example 3.11

An n-element set has 2^n subsets (without induction, cf. Theorem 2.4).
Solution. $B \mapsto$ binary string with i-th digit 1 iff $i \in B$ is a bijection (inverse: positions of the 1's); there are 2^n strings (Theorem 3.6).

Example 3.12

Ten intersections, each with a light or not, lit ones possibly with a gas station: coding A (light, station), B (light only), C (neither) gives 3^{10} possibilities.

Theorem 3.13

For $n \geq k$, the number of strings of length k over an n-element alphabet with no repeated letter is

$$(n)_k := n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

Proof. n choices for the first letter, $n-1$ for the second, …, $n-k+1$ for the k-th. ■

Example 3.14

Filling five different cabinet posts from 20 candidates: $(20)_5 = 20 \cdot 19 \cdot 18 \cdot 17 \cdot 16$ ways.

3.3 Choice problems

Motivation. Unordered selections are subsets, each counted $k!$ times among ordered ones.

How many tickets guarantee the jackpot of a lottery drawing 5 numbers from [90] (order irrelevant)?

Definition 3.15 (Binomial coefficient)

$\binom{n}{k}$ ("n choose k") denotes the number of k-element subsets of $[n]$; these are the **binomial coefficients** (Chapter 4).

Theorem 3.16

For $0 \le k \le n$: $\displaystyle\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{(n)_k}{k!}$.

Proof. Each k-subset appears $k!$ times (Theorem 3.2) among the $(n)_k$ strings of distinct elements (Theorem 3.13): $\binom{n}{k} \cdot k! = (n)_k$. ■

So the lottery needs $\binom{90}{5} = 43\,949\,268$ tickets.

Definition 3.17 (Complement)

For $S \subseteq [n]$, the **complement** S^c is the set of $i \in [n]$ with $i \notin S$.

Proposition 3.18

For $0 \le k \le n$:
(1) $\binom{n}{k} = \binom{n}{n-k}$;
(2) $\binom{n}{0} = \binom{n}{n} = 1$.

Proof. (1) $S \mapsto S^c$ is a bijection from k-subsets to $(n-k)$-subsets (inverse $T \mapsto T^c$); apply Proposition 3.10.

(2) The only 0-subset is $\emptyset$; then use (1). ■

In particular $\binom{0}{0} = 1$, and we set $\binom{n}{k} := 0$ for $k > n$.

Example 3.19

A student must work five days in January, never on two consecutive days. This can be done in $\binom{27}{5}$ ways.
Solution. For $a_1 < \cdots < a_5$ in [31], no two consecutive $\iff$ $1 \le a_1 < a_2 - 1 < a_3 - 2 < a_4 - 3 < a_5 - 4 \le 27$, so $a_i \mapsto a_i - (i-1)$ is a bijection onto the 5-subsets of [27].

Example 3.20

If the five lottery numbers from [90] are drawn *with replacement* and only the multiset counts, $\binom{94}{5}$ tickets are needed.
Solution. $f(b_1,\dots,b_5) = (b_1, b_2+1, b_3+2, b_4+3, b_5+4)$ maps multisets

$$1 \le b_1 \le b_2 \le b_3 \le b_4 \le b_5 \le 90 \tag{3.2}$$

bijectively onto the 5-subsets of [94] ((3.2) iff the shifted values increase strictly in [94]).

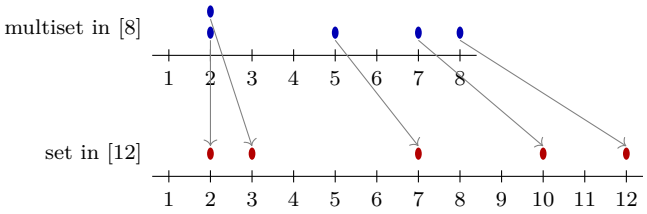


Figure. *Example 3.20 in small: adding* $0,1,2,3,4$ *sends* $\{2,2,5,7,8\}$ *to* $\{2,3,7,10,12\} \subseteq [12]$.

Theorem 3.21

The number of k-element multisets with elements from $[n]$ is $\binom{n+k-1}{k}$.

Proof [Supplement]. For $1 \le b_1 \le \cdots \le b_k \le n$ let $f(b_1,\dots,b_k) := (b_1, b_2+1, \dots, b_k+k-1)$. Since $b_i \le b_{i+1} \iff b_i + (i-1) < b_{i+1} + i$ and $b_k \le n \iff b_k + k - 1 \le n + k - 1$, f is a bijection onto the k-subsets of $[n+k-1]$ (inverse: subtract $0,\dots,k-1$). Apply Proposition 3.10. ■

	Parameters	Formula
Permutations	n distinct objects	$n!$
	a_i objects of type i, $\sum a_i = n$	$\frac{n!}{a_1!\cdots a_k!}$
Lists	k of n distinct objects	$(n)_k = \frac{n!}{(n-k)!}$
	words of length k over n letters	n^k
Subsets	k-element subsets of $[n]$	$\binom{n}{k}$
	k-element multisets from $[n]$	$\binom{n+k-1}{k}$

3.4 Selected exercises

Motivation. Book exercises: read quotients of factorials as counts, and build bijections.

Exercise 3.17 (Products of factorials)

Let $k \ge 1$ and let $b_1,\dots,b_k$ be positive integers whose sum is less than the positive integer n. Prove that $b_1!\,b_2!\cdots b_k! < n!$. Can the statement be made stronger?

Solution.

Step 1 (a count). Let $s = b_1 + \cdots + b_k \le n-1$. By Theorem 3.5, $\frac{s!}{b_1!\cdots b_k!}$ counts a nonempty set of multiset permutations, so it is a positive integer:

$$b_1!\,b_2!\cdots b_k! \le s!.$$

Step 2 (conclusion). $b_1! \cdots b_k! \le s! \le (n-1)! = \frac{n!}{n} < n!$ $(n \ge 2)$.

Stronger: the product is at most $(b_1 + \cdots + b_k)!$, with equality iff $k = 1$ (for $k \ge 2$ two types can be swapped). ■

Exercise 3.26 (Permutations without peaks)

How many permutations of $[n]$ have no entry that is larger than both of its neighbours? (The first and the last entry have only one neighbour and never violate the condition.)

Solution. Let a_n count these *peakless* permutations; we show $a_n = 2^{n-1}$.

Step 1 (where n is). For $n \ge 2$, n exceeds its neighbours, so it is the first or last entry.

Step 2 (removing n). Deleting n leaves a peakless permutation of $[n-1]$ (its neighbour becomes an end entry).

Step 3 (inserting n). Conversely, putting n at the front or end of a peakless q on $[n-1]$ gives two different peakless permutations (the new neighbour of n is smaller than n).

Step 4 (counting). So $a_n = 2a_{n-1}$, $a_1 = 1$, and $a_n = 2^{n-1}$. ■

Exercise 3.57 (Matrices with even lines)

How many $n \times n$ matrices with entries 0 and 1 have an even sum in each row and each column?

Solution. Answer: $2^{(n-1)^2}$. Call such matrices *even*; congruences are mod 2.

Step 1 (the map). $M \mapsto$ its upper left $(n-1)\times(n-1)$ block B; there are $2^{(n-1)^2}$ blocks (Theorem 3.6), so show each B extends uniquely.

Step 2 (forced entries). $m_{in} \equiv \sum_{j<n} b_{ij}$ $(i < n)$ and $m_{nj} \equiv \sum_{i<n} b_{ij}$ $(j < n)$ are forced.

Step 3 (the corner). Row n and column n both require $m_{nn} \equiv \sum_{i,j<n} b_{ij}$, so exactly one m_{nn} works.

Step 4 (conclusion). The map is a bijection (Proposition 3.10). ■